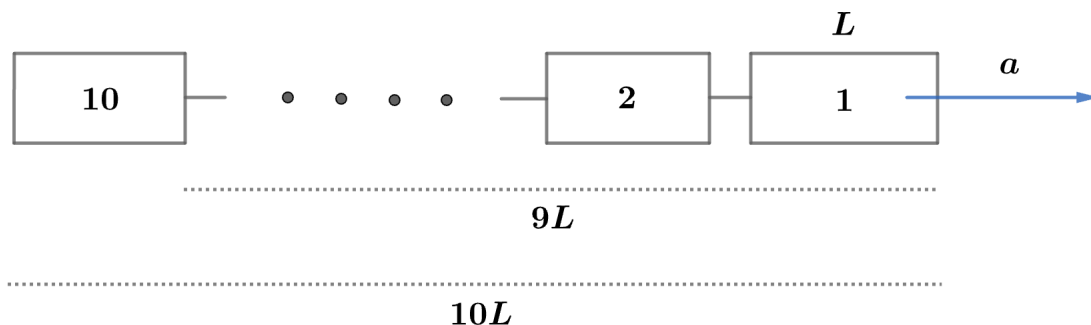


2013 F=ma Exam: Problem 1

Kevin S. Huang



Let each car have length L . If the train has acceleration a , then it takes time t_1 where

$$L = \frac{1}{2}at_1^2$$

$$t_1 = \sqrt{\frac{2L}{a}}$$

for the first car to pass the observer. Since $t \propto \sqrt{L}$, the time t_{10} it takes the first 10 cars to pass the observer is

$$t_{10} = \sqrt{10}t_1$$

Similarly, the time t_9 it takes the first 9 cars to pass the observer is

$$t_9 = \sqrt{9}t_1$$

The time it takes the 10th car to pass is the difference in times:

$$t_{10} - t_9 = (\sqrt{10} - \sqrt{9})t_1 = (\sqrt{10} - \sqrt{9})(5 \text{ s}) = 0.81 \text{ s}$$

so the answer is E.